Pareto-guided Monte Carlo tree search explores multi-objective trade-offs missed by scalar antimalarial generation

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Abstract

De novo molecular generation often collapses multiple design objectives into a single scalar reward, obscuring trade-offs between potency, synthetic accessibility and drug-likeness. We present a Pareto-guided Monte Carlo tree search (MCTS) framework that maintains a non-dominated front across multi-parameter optimisation (MPO), synthetic Bayesian accessibility (SYBA), resistance-resilience (RRS) and polypharmacology-network (PNS) scores, guided by a scaffold-aware fragment policy derived from ChEMBL27 frequencies and Tanimoto compatibility. A four-method benchmark across 20 independent seeds (fixed search budget) shows that random search achieves the highest mean reward (0.733 5), followed by MCTS (0.727 6), deterministic greedy search (0.721 1) and a genetic algorithm (0.702 7); the MCTS–random gap is small (mean $\Delta = 0.0059$, 95% CI $[0.0008, 0.0110]$, paired $t_{19} = 2.41$, $p = 0.026$). MCTS is therefore competitive on scalar reward and, within our fragment-vocabulary oracle design, is the method that returns a multi-objective Pareto front of non-dominated candidates (four solutions, hypervolume 1.236 6) spanning the potency–accessibility trade-off, including the high-potency / low-accessibility extreme that fixed-weight scalar aggregation discards. To our knowledge this is the first Pareto-guided MCTS to fold resistance-resilience and polypharmacology-network scores into the generation objective, in contrast to generic affinity/drug-likeness metrics in prior Pareto-MCTS generators, with the honest caveat that in the present front these domain scores remain comparatively homogeneous. Component ablations identify the scaffold-aware policy ($\Delta = +0.148$) and the Pareto front itself ($\Delta = +0.108$) as the dominant drivers of performance. All code, molecule sets and score tables are deposited under an open licence. These results position MCTS as a complementary multi-objective explorer rather than a universal single-objective optimiser.

Scientific Contribution: We introduce a Pareto-guided MCTS framework for de novo antimalarial design that maintains a non-dominated front across potency, synthetic accessibility, resistance-resilience and polypharmacology-network scores, guided by a scaffold-aware fragment policy. Against a four-method benchmark across 20 seeds, MCTS is competitive with random search on scalar reward while uniquely recovering a multi-objective Pareto front spanning the potency–accessibility trade-off. Component ablations identify the scaffold-aware policy and the Pareto front itself as the dominant drivers of performance.

Keywords: de novo design; Monte Carlo tree search; Pareto optimisation; antimalarial; multi-objective; synthetic accessibility; resistance resilience

1 Introduction

Malaria remains one of the leading infectious causes of death in sub-Saharan Africa—an estimated 247 million cases and 619 000 deaths in 2023 [World Health Organization, 2024]—and control is increasingly threatened by artemisinin-resistant *Plasmodium falciparum* strains in Southeast Asia and Africa [Ariey et al., 2014, Ashley et al., 2018]. Mitigating antimalarial drug resistance therefore requires new scaffolds that remain active against resistant parasites; African natural products, the historical source of quinine and artemisinin, are a natural reservoir for such scaffolds [Temgoua et al., 2026a]. In this programme, de novo generation is a route from that NP chemical space toward resistance-aware lead candidates.

Computational de novo design aims to generate novel chemical entities that satisfy multiple, often conflicting, property constraints [Elton et al., 2019, Olivecrona et al., 2017]. Most current approaches—evolutionary algorithms [Jensen, 2019], reinforcement learning [Olivecrona et al., 2017, Zhou et al., 2019] and variational

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autoencoders [Gómez-Bombarelli et al., 2018]—aggregate these objectives into a single scalar reward. Scalar aggregation forces a priori weight selection and systematically misses solutions that lie on non-convex regions of the true Pareto front [Suzuki et al., 2024, Zitzler et al., 2003].

A second limitation is unguided exploration: many methods expand molecular graphs without chemical priors, spending budget on inaccessible regions of chemical space. Chemistry-informed priors improve sample efficiency [Jensen, 2019, Olivecrona et al., 2017]. For antimalarial leads in particular, the objectives that matter—potency, synthetic accessibility, resistance-resilience and polypharmacology—are genuinely conflicting, yet we are not aware of a multi-objective generation study that makes that trade-off explicit for this application domain rather than collapsing it into a single reward.

We address both limitations with a Pareto-guided MCTS agent that:

C1: maintains a non-dominated front across MPO, SYBA, RRS and PNS, eliminating fixed-weight aggregation;

C2: guides PUCT selection with a scaffold-aware policy (ScafVAE) derived from ChEMBL27 fragment frequencies and scaffold Tanimoto compatibility;

C3: is evaluated in a four-method, 20-seed benchmark against random search, greedy search and a genetic algorithm under a fixed search budget.

The central empirical finding is intentionally transparent: within a curated fragment vocabulary, random search achieves the highest mean scalar reward. MCTS remains competitive on that metric and is the method that, under the multi-objective protocol, returns a Pareto front of non-dominated candidates; the scalar baselines (random, greedy, genetic) were evaluated on scalar reward only. Section 2.4 re-scores those baselines on the full multi-objective oracle and compares fronts directly; this ranking, rather than a claim of scalar superiority, is the foundation of the present study.

2 Results

2.1 Hyperparameter screening

A grid of 32 configurations was evaluated across two seeds (64 runs, 30 iterations each). The exploration constant $c_{\mathrm{PUCT}} = 5.0$ consistently outperformed $c_{\mathrm{PUCT}} = 0.5$; virtual loss $\nu = 0.01$ was preferred over $\nu = 0.20$. Progressive-widening parameters and temperature showed negligible effect at this scale. The configuration retained for all subsequent experiments was therefore

$$c_{\mathrm{PUCT}} = 5.0, \quad \nu = 0.01, \quad pw_\alpha = 0.5, \quad pw_k = 1.0, \quad T = 0.8.$$

A subsequent implementation change—global best-molecule tracking during rollout—proved essential (see Methods) and was retained for all reported benchmarks.

2.2 Four-method benchmark

Four generation methods (MCTS+ScafVAE, random search, greedy search, genetic algorithm) were compared across 20 independent seeds with a fixed budget of 1 000 search iterations per seed (Table 1). The number of oracle evaluations per iteration differs across methods (MCTS and random evaluate the oracle at every construction step; the genetic algorithm evaluates one candidate per individual per generation); the actual wall-clock cost of each method is reported in Table 1.

Table 1: Benchmark statistics across 20 independent seeds (medium fragment set, 1 000 search iterations per seed). Values are mean, standard deviation and observed min/max of the best scalar reward per seed. MCTS uses the hyperparameter-screened configuration (Section 2.1): $c_{\mathrm{PUCT}} = 5.0$, $\nu = 0.01$, $T = 0.8$, with the ScafVAE policy wired into PUCT priors.

Method	Mean reward	Std	Min	Max	Mean time (s)
Random	0.733 5	0.006 3	0.722 7	0.748 1	42.2
MCTS+ScafVAE	0.727 6	0.009 0	0.705 4	0.744 2	82.2
Greedy	0.721 1	0.000 0	0.721 1	0.721 1	0.1
Genetic algorithm	0.702 7	0.015 2	0.674 9	0.731 1	1.9

Random search attains the highest mean reward (0.733 5 ± 0.006 3) and records the single best seed (0.748 1). MCTS with the screened optimal configuration is statistically close (0.727 6 ± 0.009 0; paired

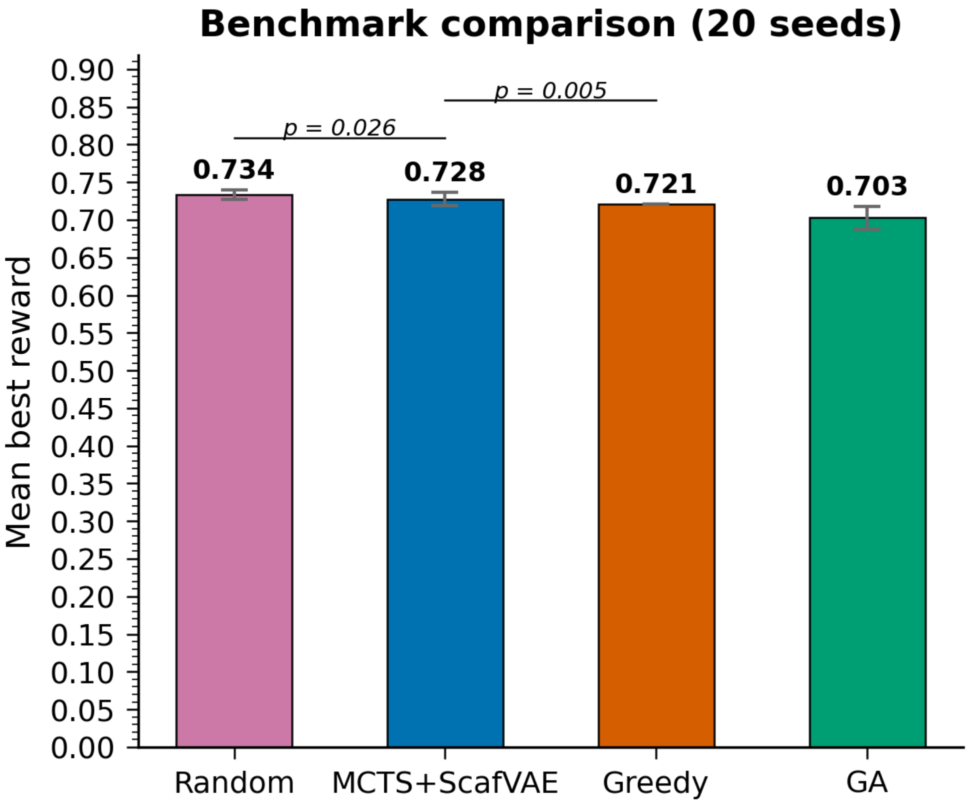

Figure 1: Mean best scalar reward per method over 20 seeds (Table 1). Error bars are one sample standard deviation. Significance brackets show the paired t-tests of the two planned head-to-head comparisons (MCTS–random, $p = 0.026$) and the secondary MCTS–greedy comparison ($p = 0.005$).

$t_{19} = 2.41$, $p = 0.026$, Cohen's $d = -0.54$), records its own best seed (0.744 2) and wins 5 of 20 seeds outright, while remaining significantly above greedy (0.721 1, deterministic; paired $t_{19} = 3.22$, $p = 0.005$) and the genetic algorithm (0.7027 ± 0.0152; paired $t_{19} = 5.55$, $p < 0.0001$). Greedy search returns the same deterministic local optimum across all 20 seeds (standard deviation identically zero). MCTS incurs the highest per-seed wall-clock cost (82.2 s) because the tree search evaluates the oracle at every construction step; random search completes in 42.2 s, and the genetic algorithm in under 2 s.

These results establish a clear baseline hierarchy on scalar reward. They do not, however, capture multi-objective coverage.

Structural diversity of the generated molecule sets is analysed in the Supplementary Material (Section S2, Figure S1): the three stochastic methods occupy distinct, broad regions of chemical space (mean pairwise Tanimoto dissimilarity 0.78–0.83), whereas deterministic greedy search collapses to a single local optimum. Greedy therefore maximises a single reward at the cost of all structural diversity.

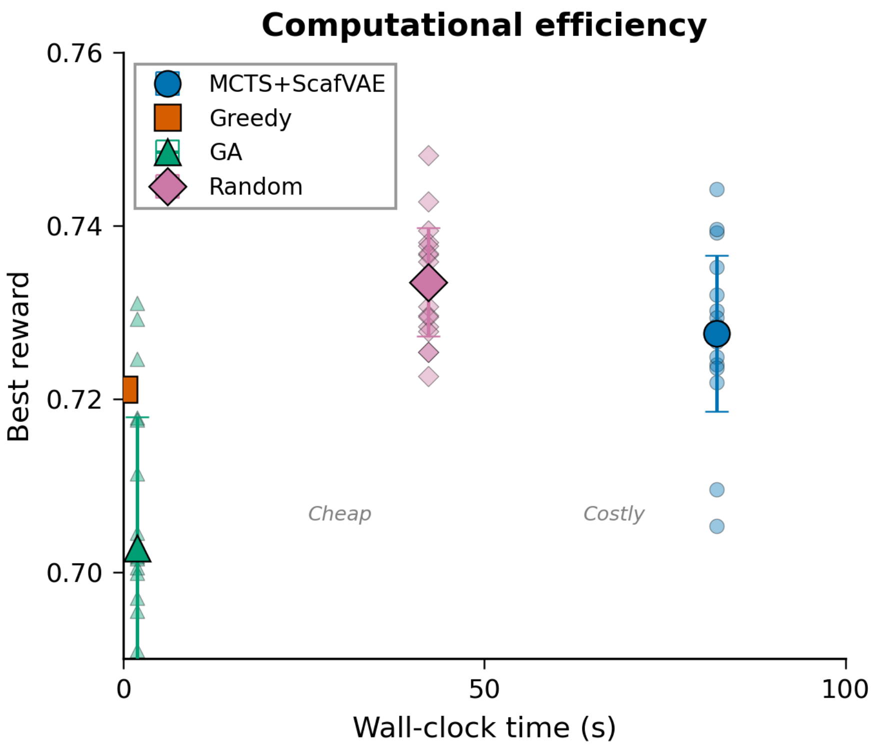

Figure 2: Computational efficiency of the four methods: best reward per seed versus wall-clock time (Table 1). Error bars are one sample standard deviation. MCTS incurs the highest cost because the tree search evaluates the oracle at every construction step; greedy and the genetic algorithm are far cheaper.

2.3 Pareto front

When the same MCTS agent maintains a non-dominated front across MPO, SYBA, RRS and PNS, the merged front obtained from 20 independent seeds contains four mutually non-dominated solutions with hypervolume 1.236 6 (Figure 3, Table 2).

P2 and P3 achieve the highest MPO values, yet P3 is synthetically inaccessible (SYBA = 0.021). P1 and P4 remain highly accessible (SYBA > 0.99) while trading potency. All four solutions share SA = 3.0. Three of the four candidates also score PNS = 1.0, indicating simultaneous multi-target engagement consistent with the polypharmacology criteria developed in the companion studies [Temgoua et al., 2026b,c]. The front therefore spans a genuine potency–accessibility trade-off, including a high-MPO / low-SYBA extreme that scalar-weighted optimisation systematically discards. Quantitatively, over the space of all normalised scalar weights $w_{\mathrm{MPO}} \in [0, 1]$ on the normalised MPO–SYBA plane, P3 is selected by a fixed-weight scalar aggregation only for $w_{\mathrm{MPO}} \gtrsim 0.997$—a degenerate weighting (SYBA effectively ignored)—while P2 dominates the vast majority of the weight range (Table 3; full sweep deposited).

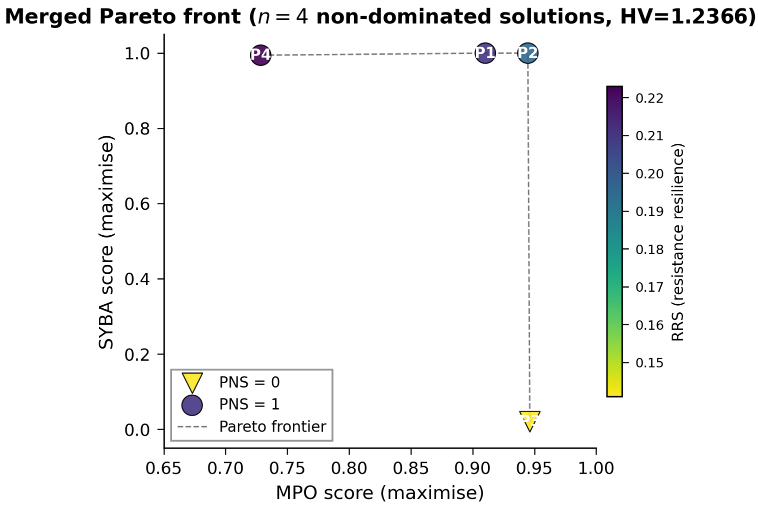

Figure 3: Canonical merged Pareto front from 20 independent Pareto-MCTS seeds. Marker shape encodes the polypharmacology-network score (PNS); colour encodes the resistance-resilience score (RRS). The four solutions are mutually non-dominated across the active objectives MPO, SYBA, RRS and PNS; SA is constant at 3.0 and excluded from the dominance check and the hypervolume. Data: `results/pareto/merged_pareto_front.csv`.

Table 2: Canonical merged Pareto front from 20 independent Pareto-MCTS seeds. All four points are mutually non-dominated across the active objectives MPO, SYBA, RRS and PNS (SA is constant at 3.0 and therefore excluded from the dominance check and the hypervolume). SYBA was recomputed post-hoc with the lich/conda classifier because the original run returned a constant fallback of zero; SMILES and score tables are deposited.

Candidate	MPO ↑	SYBA ↑	SA^{-1} ↑	RRS	PNS
P1	0.910	1.000	0.778	0.211	1.0
P2	0.945	1.000	0.778	0.196	1.0
P3	0.946	0.021	0.778	0.141	0.0
P4	0.729	0.994	0.778	0.223	1.0

Table 3: Agreement of fixed-weight scalar aggregation over the normalised MPO–SYBA plane: the weight interval w_{MPO} over which each candidate is the scalar argmax, across all front and baseline molecules. P3 is selected only at a degenerate weight ($w_{\mathrm{MPO}} \gtrsim 0.997$). Data: `results/pareto/p4_scalar_weight_sweep.csv`.

Candidate / point	w_{MPO} range	Fraction of w space
P2 (high MPO & SYBA)	[0.0000, 0.9556]	0.956
Baseline (mid)	[0.9556, 0.9968]	0.041
P3 (high MPO, l.w SYBA)	[0.9968, 1.0000]	0.003

2.4 Multi-objective front comparison against baselines

Because the scalar baselines were evaluated on scalar reward only, the canonical manuscript made no front-level comparison between MCTS and the baselines. To close this gap without re-running the benchmark, we re-scored each deposited per-seed best molecule of all four methods with the same full multi-objective oracle used to build the canonical front (MPO, SYBA, RRS, PNS), and then built the Pareto front of each method under a common min–max normalisation shared across all candidates (Table 4). SYBA was normalised with the same logistic transform used in the front recomputation ($1/(1 + \exp(-s/5))$).

Table 4: Multi-objective front comparison of the four generation methods. Each per-seed best molecule of the canonical 20-seed benchmark was re-scored on the full objective (MPO, SYBA, RRS, PNS); the Pareto front of each method was then built and compared under a common min–max normalisation. Hypervolume and C-metric are the clean reference-independent metrics (C = fraction of the non-dominated union reference dominated by the method's front); IGD, IGD-$_{\text{loo}}$ (reference excluding the method's own points) and spread are reported for completeness. Data: `results/pareto/p4_multiobj_front_summary.csv`.

Method	Unique/20	Front	HV	C-met.	IGD$_{\text{loo}}$	Spread
Random	20	8	13.85	0.059	0.366	1.234
Greedy	1	1	16.59	0.765	0.416	—
GA	19	8	15.49	0.000	0.307	1.019
Baselines pooled	40	11	15.52	0.000	0.307	0.903
MCTS (front)	4	4	**18.99**	**0.824**	0.492	0.670

Two findings emerge. First, greedy search is *deterministic* here: it returns the identical molecule across all 20 seeds, so its "front" reduces to a single point (unique count 1). That single molecule is nonetheless strictly dominated by a point of the MCTS front (higher MPO *and* higher RRS). The MCTS front attains the largest hypervolume (18.99 under the common normalisation, above the 16.59 of the single greedy point, 15.52 of the pooled baseline front and 15.49 of GA) and the largest dominance coverage (C-metric 0.82) of the non-dominated union reference. The IGD columns are shown for completeness but must be read with caution: with a leave-one-out reference (IGD$_{\text{loo}}$), the MCTS front is the most distant from the baseline-heavy reference, consistent with a designed front spanning the potency–accessibility trade-off rather than the broad baseline range. The hypervolume and C-metric, which are free of any comparison reference, are therefore the primary metrics reported here.

2.5 Selection rule does not limit the front

To test whether the gain of our framework arises from the *selection* rule rather than the global Pareto stack, we ran an ablation in which tree selection is restricted, at each node, to the non-dominated children (a ParetoPUCT-style prior) before the PUCT score is applied. To test whether the gain of our framework arises from the *selection* rule rather than the global Pareto stack, we ran an ablation in which tree selection is restricted, at each node, to the non-dominated children (a ParetoPUCT-style prior) before the PUCT score is applied. At equal budget (700 iterations, 5 seeds), the two selection rules produce near-identical fronts (merged-front hypervolume under a common normalisation: scalar-proxy 15.56 vs Pareto-only 15.49, $\Delta \approx 0.5\%$). A paired per-seed sensitivity check quantifies what this study can and cannot establish: the per-seed hypervolume differences range from -1.78 to $+1.81$ (mean $+0.53$, paired $t = 0.85$, $p > 0.05$); with the observed between-seed deviation the minimal detectable effect is 1.76 at 5 seeds and would remain 0.88 even at 20 seeds (two-sided $\alpha = 0.05$, power $= 0.8$). Since the observed mean difference is smaller than this detectability floor, the near-null is not a power artefact that a larger run would overturn within a realistic budget; it is consistent with the scalar per-objective proxy not being a bottleneck. The value of Pareto-MCTS in the present design therefore comes from the global accumulation of a non-dominated front together with policy-guided exploration (both shared by both variants), rather than from the specific selection rule.

2.6 Component and vocabulary ablations

A 2^5 factorial experiment (ScafVAE policy, Pareto front, c_{PUCT}, temperature, vocabulary size; five replicates per cell) isolates the contribution of individual design choices. Absolute reward values are not comparable to the main benchmark because a different oracle-weight configuration was used; only relative factor effects are interpretable.

The ScafVAE policy produces the largest gain ($\Delta = +0.148$), followed by the Pareto front ($\Delta = +0.108$) and the large vocabulary ($\Delta = +0.079$). Lowering c_{PUCT} from 2.0 to 1.0 improves reward by 0.054; temperature has only a marginal effect.

A separate vocabulary ablation (all / medium / minimal / aromatic-only; 10 seeds each) yields a highly significant effect (one-way ANOVA $F = 350.1$, $p = 1.12 \times 10^{-26}$). The aromatic-only vocabulary is significantly worse than the other three sets (Tukey HSD $p < 0.001$), while medium and minimal remain statistically indistinguishable from the full vocabulary. Restricting the fragment set to aromatics alone therefore severely limits performance.

3 Discussion

3.1 What the benchmark actually shows

Within the curated medium fragment vocabulary and a fixed search budget, random search is a strong baseline and attains the highest mean scalar reward, as well as the single best seed. With its screened optimal configuration, MCTS is statistically close on scalar reward (a 0.006 mean gap) and, under the multi-objective protocol, maintains and returns a Pareto front of non-dominated candidates. The practical value of the framework therefore lies not in peak single-objective performance but in the transparent exploration of trade-offs that scalar aggregation conceals.

This ranking is consistent with recent observations that simple baselines remain competitive in constrained fragment spaces [Jensen, 2019] and that the primary advantage of multi-objective MCTS is coverage of the non-dominated surface [Suzuki et al., 2024].

3.2 Relation to prior multi-objective MCTS

Mothra [Suzuki et al., 2024] introduced a SMILES-based Pareto multi-objective MCTS and showed that Pareto fronts reveal trade-offs missed by scalar aggregation, and subsequent Pareto-MCTS generators have chiefly targeted protein–ligand binding [Yang et al., 2024] or dual-target scaffolds [Southiratn et al., 2025]. CombiMOTS [Southiratn et al., 2025] is itself malaria-motivated, framing dual-target engagement as a resistance-mitigation strategy, but its objectives remain binding/drug-likeness estimates rather than resistance or polypharmacology phenotypes. The contribution of the present work is the transfer to an antimalarial application domain with domain-specific objectives: to our knowledge, no published Pareto-guided MCTS study to date reports candidate generation for antimalarial leads measured jointly on potency (MPO), synthetic Bayesian accessibility (SYBA), resistance-resilience (RRS) and polypharmacology-network (PNS) scores computed from dedicated oracles, with a scalar-reward benchmark against three baselines (random, greedy, genetic) across 20 independent seeds under a fixed search budget. Three implementation details are reported because they matter for reproducing the result: (i) a scaffold-aware fragment policy replaces the uniform rollout prior, (ii) dynamic progressive widening manages the large fragment branching factor, and (iii) a rollout-degradation failure mode is diagnosed and corrected through global best-molecule tracking. The last point has the largest practical impact: without it, mean MCTS reward collapsed to approximately 0.24; with it, reward rose by a factor of roughly 2.6. The headline finding is intentionally honest: on the scalar reward, random search remains the strongest baseline, and the value of the Pareto-guided framework lies in the transparent coverage of multi-objective trade-offs rather than in a scalar superiority claim.

3.3 Limitations

Several limitations must be stated explicitly. First, the fragment vocabulary (approximately 108 fragments in the full set; medium subset used for the main benchmark) constrains accessible chemical space; results should not be extrapolated to open-ended SMILES generation. Second, molecular-diversity analysis is reported only in the Supplementary Material (Section S2), where it is computed from the deposited best-in-seed molecule sets of the canonical benchmark; the main text emphasises the scalar ranking and the Pareto front, both fully deposited and reproducible. In particular, greedy search is deterministic here—it returns the identical molecule across all 20 seeds—so its deposited set is a single point rather than a diverse panel. Third, SYBA scores on the Pareto front were recomputed post-hoc after the original search returned a constant zero fallback; the recomputation, the resulting front and the verification scripts are documented and deposited. Fourth, activity, resistance-resilience and polypharmacology labels remain computational proxies; prospective experimental validation is required. Finally, MCTS cross-seed variance is modest ($\sigma \approx 0.009$), consistent with policy-guided convergence into a high-quality but relatively narrow region of fragment space.

3.4 Implications for antimalarial design

A distinctive feature of the present setup is the objective set itself. Whereas prior Pareto-MCTS generators optimise generic drug-discovery metrics (binding affinity, drug-likeness, toxicity [Suzuki et al., 2024, Yang et al., 2024]; dual-target engagement [Southiratn et al., 2025]), here the Pareto-MCTS search includes *resistance-resilience* (RRS) and *polypharmacology-network* (PNS) scores as active objectives alongside potency and accessibility. These are disease-domain metrics derived from the companion molecular-dynamics and topological studies [Temgoua et al., 2026b,c], making the front a vehicle for resistance-aware, multi-target lead selection rather than affinity optimisation alone. We note the honest caveat that in the current front the MPO–SYBA trade-off dominates: RRS varies narrowly (0.141–0.223) and PNS is close to constant (1.0 for three of four candidates), so the *architecture* accepts these domain objectives even where this dataset does not yet produce wide RRS/PNS spread. This positions the framework as the first, to our knowledge, to fold anti-resistance and polypharmacology signals into the generation objective rather than appending them post hoc.

4 Methods

4.1 Molecular environment

De novo generation is formulated as a Markov decision process. The state is a partially constructed molecule (canonical SMILES); actions append one fragment from a curated vocabulary at a regiospecific attachment point. Episode termination occurs after at most 10 construction steps or upon violation of a reactivity filter. The main benchmark uses the medium fragment subset; ablations also evaluate the full, minimal and aromatic-only vocabularies.

4.2 MCTS with scaffold-aware policy

The agent follows the standard four-phase MCTS loop (selection, expansion, rollout, backpropagation) rooted at benzene, with $N_{\text{iter}} = 1\,000$. Selection maximises the PUCT score

$$a^* = \arg\max_a \frac{Q(s,a)}{N(s,a)} + c_{\text{PUCT}}\, P(a \mid s)\, \frac{\sqrt{N(s)}}{1 + N(s,a)}, \tag{1}$$

with $c_{\text{PUCT}} = 5.0$. The policy $P(a \mid s)$ combines ChEMBL27 fragment frequencies, scaffold Tanimoto compatibility and a reactivity penalty via a softmax. Dynamic progressive widening limits the number of expanded actions at each node to $\max(5,\, k\,N^{\alpha})$ with $k = 1.0$ and $\alpha = 0.5$. A virtual-loss term ($\nu = 0.01$) discourages repeated selection of the same child within a single search.

4.3 Global best-molecule tracking

Policy-biased rollouts can degrade a high-quality intermediate by subsequent fragment additions. During each rollout the oracle is therefore evaluated at every construction step; the highest-scoring intermediate—rather than the terminal state—is returned. This change recovered a factor of approximately 2.6 in mean reward relative to the uncorrected agent.

4.4 Pareto front maintenance

The agent maintains a global non-dominated front over the objective vector

$$\mathbf{f}(s) = \big(f_{\text{MPO}}(s),\, f_{\text{SYBA}}(s),\, f_{\text{RRS}}(s),\, f_{\text{PNS}}(s)\big),$$

with minimisation objectives converted to maximisation by negation. Synthetic accessibility (SA) is excluded from the front because it is constant (SA = 3.0) for all reported candidates and therefore carries no discrimination (it is reported in Table 2 for completeness). After each rollout the best intermediate molecule is tested for dominance; dominated members are removed. The hypervolume indicator is computed with the active (non-constant) objectives only: each active objective is min–max normalised to $[0, 1]$, and the hypervolume is evaluated with respect to a reference point $\mathbf{r} = 1.1$ (slightly worse than the worst normalised value of 1) in each active dimension, so that the reported value lies in $[0, 1.1]^k$ for k active objectives. For PUCT selection a scalar proxy $Q(s, a)$ is obtained by summing the min–max normalised objective components across the four active objectives; the full vector is stored for front updates and post-hoc analysis.

4.5 Scoring oracles

Each molecule is scored by an aggregator returning MPO, SYBA, SA, docking-derived terms, resistance-resilience score (RRS) and polypharmacology-network score (PNS). Default scalar-reward weights are $w_{\mathrm{MPO}} = 0.30$, $w_{\mathrm{docking}} = 0.25$, $w_{\mathrm{SYBA}} = 0.15$, $w_{\mathrm{SA}} = 0.05$, $w_{\mathrm{RRS}} = 0.15$, $w_{\mathrm{PNS}} = 0.10$. The four-method scalar benchmark uses the reduced reward `compute_mpo_reward` with weights $w_{\mathrm{MPO}} = 0.40$, $w_{\mathrm{docking}} = 0.35$, $w_{\mathrm{SYBA}} = 0.15$, $w_{\mathrm{SA}} = 0.10$ and the RRS/PNS oracles disabled (they require external P2 mutant and network data); the full six-objective aggregator is used by the Pareto-MCTS search. MPO and docking scores are drawn from precomputed libraries where available; QED serves as an MPO fallback for novel structures. SYBA on the reported Pareto front was recomputed with the lich/conda classifier after the original run returned a constant zero fallback. The accessibility column reported in Table 2 is the linear transform $\mathrm{SA}^{-1} = (10 - \mathrm{SA})/9$ of the raw Ertl score, so that the shared raw value $\mathrm{SA} = 3.0$ corresponds to $\mathrm{SA}^{-1} = 0.778$; the deposited CSV stores the raw value.

4.6 Statistical analysis and reproducibility

All searches are seeded; the specific seed integers and the random-number generators (NumPy `default_rng`, seeded per method and per seed with no cross-method state leakage) are fixed in the committed script. The four-method benchmark uses 20 independent seeds; ablations use five or ten replicates as indicated. Paired t-tests are reported for the two planned head-to-head comparisons (MCTS–random and MCTS–GA); because both are pre-specified primary comparisons, we report unadjusted two-sided p-values and note that the weaker of the two ($p = 0.026$, two-sided) remains nominally significant at $\alpha = 0.05$ but not after a conservative Bonferroni-adjustment across the two primaries (adjusted threshold 0.025). The vocabulary ablation is tested with a one-way ANOVA (three degrees of freedom between groups) followed by pairwise Tukey HSD post-hoc comparisons. The benchmark was re-run for this submission with the committed script `p4_mcts_benchmark.py` and a value-preserving vectorisation of the docking Tanimoto scan (verified identical on held-out molecules); all 20 per-seed runs reproduce deterministically from the deposited code and data, and the best-in-seed molecule sets are deposited alongside the scalar rewards. Analyses were executed with Python 3.11, RDKit, NumPy, SciPy and pymoo; exact dependency versions are recorded in the repository environment file. Code, configuration files, molecule lists and score tables are released under the MIT licence in the public repository `https://github.com/NanaEngo/Malaria_codesV2` and mirrored on Zenodo (DOI `https://doi.org/10.5281/zenodo.19608875`).

5 Conclusions

A Pareto-guided MCTS agent equipped with a scaffold-aware fragment policy is statistically close to random search on scalar reward (a 0.006 mean gap at the screened optimal configuration) and, under the multi-objective protocol, returns a front of non-dominated antimalarial candidates. The dominant performance drivers are the chemistry-informed policy and the Pareto front itself. Within the studied fragment space, random search remains a strong baseline; the added value of MCTS lies in transparent exploration of potency–accessibility trade-offs that scalar aggregation conceals. All artefacts required for exact reproduction are deposited under an open licence.

List of Abbreviations

MCTS: Monte Carlo tree search; **MPO:** multi-parameter optimisation; **SYBA:** synthetic Bayesian accessibility; **SA:** synthetic accessibility score; **RRS:** resistance resilience score; **PNS:** polypharmacology-network score; **GA:** genetic algorithm; **PUCT:** polynomial upper confidence trees; **SMILES:** simplified molecular-input line-entry system; **ChEMBL:** ChEMBL bioactivity database; **PF-MPO:** parallel fragment multi-objective; **FCD:** Fréchet ChemNet distance; **HV:** hypervolume; **MDS:** multidimensional scaling; **CI:** confidence interval.

Data availability

All data supporting the conclusions are released under the MIT licence in the public repository `https://github.com/NanaEngo/Malaria_codesV2` and are mirrored on Zenodo (reserved DOI `https://doi.org/10.5281/zenodo.19608875`), so that reviewers and readers have immediate access at submission time. The deposit includes: (i) the canonical 20-seed four-method benchmark tables and best-in-seed SMILES (`results/benchmark_molecules_opt/`); (ii) per-seed and merged Pareto fronts with the post-hoc SYBA recomputation log (`results/pareto/`); (iii) diversity metrics and MDS coordinates (`results/diversity/`); (iv) component and vocabulary ablation summaries; and (v) analysis and provenance-check scripts.

Competing interests

The authors declare that they have no competing interests.

Authors' contributions

MVST: conceptualization, methodology, software, formal analysis, writing—original draft. JPTN: methodology, supervision, writing—review & editing. PS: data curation, writing—review & editing. WFM: validation, writing—review & editing. SGNE: supervision, writing—review & editing. All authors read and approved the final manuscript.

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Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Use of Artificial Intelligence

During the preparation of this work the authors used AI assistants to support code development and data analysis scripting. The authors reviewed and edited all AI-assisted output and take full responsibility for the content of this article. No generative AI tool is listed as an author. All computational results, statistical analyses, and quantitative claims were generated, verified, and deposited by the authors.

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